

APARRI, Philippines

WMO: 982320

Lat: 18.3604N

Lon: 121.6304E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.8	70.4	64.9	92.1	71.2	66.1	96.3	71.6	27.8	72.7	26.5	73.1	11.3	40	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.4	92.6	83.3	91.5	83.1	90.6	82.7	84.4	90.7	83.7	90.0	83.4	89.5	7.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.6	170.1	89.3	81.9	166.1	88.8	81.6	164.2	88.5	48.4	90.8	47.9	90.7	47.4	90.0	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.5	22.7	19.8	DB	65.6	97.9	2.9	2.9	63.5	100.0	61.8	101.7	60.2	103.4	58.0	105.5
			WB	62.9	86.5	2.7	2.1	61.0	88.0	59.4	89.2	57.9	90.4	55.9	92.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	75.9	76.8	79.7	82.7	84.2	85.0	84.0	83.7	83.2	82.0	80.0	77.2	(6)
	DBStd	3.81	2.69	2.98	2.81	2.47	2.01	1.78	2.11	1.95	1.76	1.93	2.18	2.59	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11394	803	751	922	980	1059	1051	1055	1044	996	992	899	842	(10)
	CDD65	5919	338	331	457	530	594	601	590	579	546	527	449	377	(11)
	CDH74	60732	1757	2117	3999	5853	7148	7531	7093	6823	6300	5657	4146	2308	(12)
	CDH80	19697	130	273	927	1997	2873	3334	2823	2567	2215	1585	763	210	(13)
(14) Wind	WSAvg	7.8	8.7	8.4	8.2	7.1	6.4	6.1	6.4	6.6	6.8	9.0	9.2	10.2	(14)
(15) Precipitation	PrecAvg	83.1	4.6	2.8	1.8	1.3	4.3	5.1	7.7	8.6	8.9	12.6	12.8	8.1	(15)
	PrecMax	120.7	16.5	13.6	11.0	5.7	12.4	16.5	27.4	24.3	25.2	32.3	52.8	19.5	(16)
	PrecMin	55.7	0.1	0.0	0.0	0.0	0.0	0.1	0.5	0.7	0.4	2.0	1.2	0.9	(17)
	PrecStd	20.9	3.6	2.6	2.2	1.4	3.1	3.8	6.3	5.4	5.2	6.7	10.7	4.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	87.1	90.0	92.9	93.5	93.8	93.5	92.4	91.4	90.1	88.4	86.7	(19)
		MCWB	78.6	81.0	81.7	83.2	83.5	83.8	83.6	83.4	83.3	82.9	81.8	80.8	(20)
	2%	DB	82.7	84.9	87.9	90.5	91.7	92.5	91.7	90.9	89.8	88.5	86.5	83.9	(21)
		MCWB	77.9	79.0	81.3	82.4	82.9	83.4	83.1	83.1	82.5	82.1	80.9	79.2	(22)
	5%	DB	81.6	83.1	86.3	89.1	90.5	91.4	90.5	89.7	88.8	87.2	85.1	82.2	(23)
		MCWB	77.2	77.8	80.2	82.0	82.6	83.1	82.7	82.8	82.0	81.4	80.1	78.0	(24)
	10%	DB	80.3	81.6	84.8	87.7	89.2	90.3	89.1	88.5	87.7	86.0	83.8	80.8	(25)
		MCWB	76.2	77.0	79.4	81.3	82.1	82.7	82.2	82.2	81.5	80.6	79.3	77.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	81.5	83.2	84.1	84.6	85.5	84.6	84.5	84.2	83.6	82.7	81.6	(27)
		MCDB	83.5	86.1	88.9	90.5	91.0	91.1	90.9	90.3	90.0	89.0	87.2	85.7	(28)
	2%	WB	78.5	79.8	81.8	83.3	83.7	84.3	83.8	83.7	83.3	82.6	81.4	79.5	(29)
		MCDB	82.1	84.3	87.1	89.5	90.1	90.9	90.3	89.6	88.9	87.8	85.9	83.2	(30)
	5%	WB	77.4	78.5	80.8	82.5	83.1	83.6	83.2	83.1	82.6	81.8	80.4	78.3	(31)
		MCDB	80.9	82.5	85.8	88.5	89.5	90.2	89.4	88.9	88.0	86.6	84.6	81.6	(32)
	10%	WB	76.4	77.3	79.7	81.6	82.4	83.0	82.5	82.4	81.8	80.9	79.5	77.3	(33)
		MCDB	79.8	81.0	84.3	87.2	88.6	89.4	88.5	88.0	87.0	85.4	83.4	80.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.5	8.8	10.1	10.8	11.4	10.4	9.7	9.5	8.2	7.1	5.9	(35)
		MCDBR	8.9	10.0	11.4	12.0	12.4	12.6	12.3	11.5	11.0	9.9	9.2	8.6	(36)
	5% WB	MCWBR	6.6	7.1	7.7	7.6	7.2	7.2	7.0	7.0	7.0	6.5	6.7	5.9	(37)
		MCWBR	8.4	9.0	11.0	11.6	11.9	12.1	11.7	11.0	10.5	9.6	8.9	8.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.438	0.435	0.494	0.495	0.464	0.447	0.440	0.446	0.459	0.474	0.472	0.482	(40)
		taud	2.325	2.352	2.175	2.201	2.330	2.391	2.409	2.381	2.341	2.263	2.252	2.181	(41)
	Ebn at Noon		267	275	262	261	266	268	271	271	268	259	254	248	(42)
		Edh at Noon	38	39	48	47	41	38	38	39	40	43	41	43	(43)
(44) All-Sky Solar Radiation	RadAvg		1066	1345	1592	1914	1894	1899	1739	1612	1578	1347	1132	899	(44)
	RadStd		105	197	154	149	126	148	160	129	120	104	154	123	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	-0.70	-0.59	-0.45	N/A	N/A	N/A	N/A

Nomenclature: See separate page